

Bar inventory forecasting: full project explanation

A companion guide | 28 September 2026

This guide explains the why, math, code flow, test results and interview talking points. The separate two-page report.pdf is the form submission document. These are historical examples, not live stock or purchase orders.

1. What is the business decision?

The hotel operates six bars and wants to avoid two failures: stockouts of high-demand items and cash tied up in slow stock. Every morning a manager needs a forecast of usage, a target stock level (par), and a suggested order quantity for each bar-brand. Brands are the operational item identifier in the supplied sheet; if the same brand has different bottle sizes or SKUs, production must split them. All quantities here use millilitres. Nothing automatically orders stock.

2. What is in the dataset?

The supplied spreadsheet has 6,575 rows dated from 1 January 2023 through 1 January 2024, six bars, 16 brands, and 96 bar-brand combinations. Columns are timestamp, bar, alcohol type, brand, opening balance, purchase, consumed and closing balance. There are no null values and no exact duplicate records. There are 86 rows where opening + purchase - consumed differs from closing by more than 1 ml. There are 129 transitions where a later opening differs from the previous logged closing by more than 1 ml. These need reconciliation with receipts, spoilage, transfers or data entry before operations.

3. Why time between records matters

A bar-brand is logged once per observed date in this file, but the gaps between logs vary; the median gap is 3.86 days and the 90th percentile is 11.27 days. The notebook assumes the consumed figure reflects usage since that bar-brand's prior record. If 300 ml is recorded after three days, the estimated daily rate is 100 ml/day. Filling the two unlogged days as zero would create fake sales data. The first row for each of the 96 groups is excluded from forecasting evaluation because there is no known start date for that consumption interval. This interpretation is plausible because most openings line up with prior closings, but the hotel must verify the logging semantics. If it is actually point-in-time usage, this method must change.

4. Data cleaning and leakage prevention

Parse timestamps as month/day/year hour:minute; sort within bar-brand; calculate elapsed_days from the preceding timestamp; retain positive durations and nonnegative consumption. Calculate interval rate = consumed_ml / elapsed_days. For extreme rate anomalies, clip the history rate to the 99th percentile of the TRAINING period only (709.8 ml/day). Never estimate a cap from the whole dataset and then claim an untouched test. Accounting differences are displayed rather than silently repaired. The notebook keeps those rows in the prototype because blindly dropping them could bias the sample; a real deployment must reconcile them.

5. Forecast models and choice

The target for an observed interval is total consumed millilitres. Three transparent rates are compared: (A) lifetime: total past consumed / total past elapsed days; (B) recent-six: total of six previous intervals / their elapsed days; (C) shrunk-six: mix recent rate and lifetime rate with a 30-day prior weight. Each prediction multiplies the rate by the target interval duration. Only previous rows enter that prediction. This is rolling one-step validation: after an actual event arrives, it becomes available for the next prediction. It is not a claim that one fixed forecast predicts the entire test period in advance. A weighted lifetime average handles irregular gaps better than an average of row rates. More complex seasonal, tree and neural models are hard to defend with sparse, uneven logs and no occupancy or stockout labels; they may become useful with richer daily data.

6. Chronological evaluation and honest result

The training window ends before 1 October 2023. Validation covers 1 October to 14 November, and held-out testing starts 15 November 2023 and ends 1 January 2024. Select the lowest validation mean absolute error, not the model with the best test score. Validation MAE per logged interval: lifetime 245.79 ml, shrunk-six 248.38 ml, recent-six 276.30 ml. Lifetime wins. On 852 held-out intervals its MAE is 247.80 ml, WAPE is 82.97%, and mean actual-minus-prediction bias is -1.74 ml. MAE means a typical absolute miss of about 248 ml per event interval. WAPE divides total absolute error by total actual usage. A near-zero aggregate bias can still hide large item-level errors. These results are weak and do not prove that the par policy prevents stockouts.

7. Convert a daily rate into par

Par is the inventory target to cover supplier lead time and time until the next review, plus a risk buffer. Formula: $\text{ceil}[(\text{predicted ml/day} \times (\text{lead days} + \text{review days}) \times (1 + \text{buffer})) / \text{rounding step}] \times \text{rounding step}$. The prototype assumes 2-day lead, 1-day review, 20% buffer and 100 ml rounding. Example: $79 \text{ ml/day} \times 3 \times 1.20 = 284.4 \text{ ml}$, rounded to 300 ml par. These are illustrative inputs, not learned from the sheet, not a calibrated 95% service guarantee, and not necessarily valid bottle pack sizes. Once a live physical count and open purchase orders are available, suggest $\text{max}(0, \text{par} - \text{current on-hand} - \text{on-order})$, rounded to a supplier pack size. The notebook assumes open orders = zero solely because there is no column for them, and uses the last logged closing balance to illustrate the arithmetic. The data is over two years old as of this guide: never use its suggested quantities for real ordering.

8. Understand the seven-day simulation

The simulation selects the highest estimated daily-demand group: Taylor's Bar / Grey Goose, about 79 ml/day, par 300 ml. A made-up starting physical count of 150 ml is used. At each daily review, outstanding orders and on-hand stock make the inventory position. An order placed today arrives after two days. The scenario requests predicted daily demand, with 1.5 times demand on days five and six, and tracks receipts, orders, served, unserved and ending stock. The example has about 8 ml unmet demand on day two despite orders. This is a useful demonstration of the lead-time risk, but its numbers are synthetic; there is no observed benefit claim or historical counterfactual. To estimate business impact, replay actual daily sales and lead times and compare with the hotel's existing rule.

9. What happens in a real hotel?

A nightly data pipeline ingests POS sales, physical counts, purchases, delivery confirmations, waste, breakage and transfers for every bar and SKU. The forecast updates. At a scheduled review, the manager sees demand estimate, uncertainty, par, current stock, incoming stock, proposed quantity, freshness of data and any bad-data warnings. Supplier minimums, pack

sizes, shelf life, budget and service target are applied before a human approves any purchase order. Keep an audit trail of overrides. Do not turn a model suggestion into an automatic spend without explicit business authorization.

10. Failure modes and next iteration

Consumption is not the same as true demand during stockouts; a 0 ml sale could mean no customers or no stock. Irregular intervals can mix multiple seasons or events; inventory records can omit transfers. Empty counts, stale POS feeds, renamed brands, and unrecorded pending orders can all distort par. Start by collecting daily true sales, lost-sale/stockout indicators, stock counts and actual supplier lead times. Backtest a fixed daily forecast horizon and compare naive seasonal baselines and current ordering rules. Calibrate uncertainty or quantile forecasts to service-level and holding-cost targets rather than assuming 20%. Monitor MAE, WAPE and bias by item, stockout days, days of cover, spoilage, stale data, lead-time drift and human overrides. New items need pooled or category-level fallbacks.

11. How to navigate and explain the notebook

Cell 1 loads the CSV. Cell 2 audits balances, gaps and plots. Cell 3 builds interval rates and chronological splits. Cell 4 walks forward through three model choices and reports held-out metrics. Cell 5 creates 96 illustrative par recommendations and exports CSV. Cell 6 demonstrates daily reorder logic with a synthetic scenario. The final Markdown cell is the limitations and next-step handoff. Use `demo_walkthrough.md` as a video outline, but explain it in your own voice and demonstrate the actual notebook outputs. The form asks for a separate 3-5 minute recording and a 1-2 page PDF report; do not submit this long guide instead of the short report.

Key terms in plain English

- Bar-brand: One brand at one bar; the forecast and par are specific to that pair.
- Interval: Time between two logged rows for the same bar-brand.
- MAE: Average absolute amount by which predictions missed actual logged consumption, in ml per logged interval.
- WAPE: Total absolute prediction error divided by total actual usage, expressed as a percentage.
- Par level: Target quantity to maintain to cover expected demand while waiting for replenishment.
- Inventory position: On-hand stock plus confirmed stock on order, less any allocated/backordered demand.

Data source:

https://docs.google.com/spreadsheets/d/14i7oWnBOolf3pai37bsGYkby3VXJL1AIX_KxC8ZMKtg/edit?gid=1742513311#gid=1742513311